# Ecommerce_Sales_Analysis

Database Design & SQL Query Project

*Prepared by Basheerunnisa*

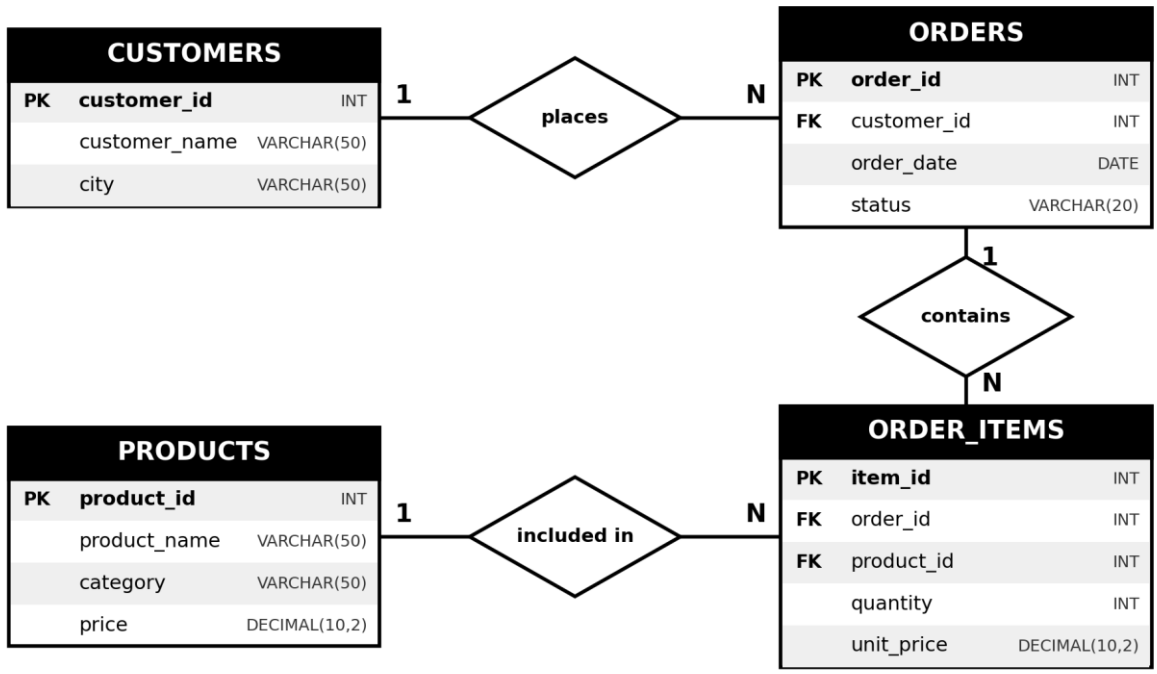
## Theme

An online store wants to maintain customer details, product catalogue, orders and the items sold in each order, so that sales can be analysed by product, customer, city and order status.

## Modules

- **Customers** – customer name and city
- **Products** – product name, category and price
- **Orders** – order date and delivery status for each customer
- **Order Items** – products, quantity and unit price in each order

## ER Diagram




ER Diagram – Ecommerce Sales Analysis (PK = Primary Key, FK = Foreign Key)

### Relationships

- **Customers → Orders:** One customer can place many orders (1 : N).
- **Orders → Order Items:** One order can contain many order items (1 : N).
- **Products → Order Items:** One product can appear in many order items (1 : N).

Order Items is the bridge table that makes Orders and Products a many-to-many relationship.

# Table Structure

### CUSTOMERS

| Column | Data Type | Key |
|---|---|---|
| customer_id | INT | Primary Key |
| customer_name | VARCHAR(50) | |
| city | VARCHAR(50) | |

### PRODUCTS

| Column | Data Type | Key |
|---|---|---|
| product_id | INT | Primary Key |
| product_name | VARCHAR(50) | |
| category | VARCHAR(50) | |
| price | DECIMAL(10,2) | |

### ORDERS

| Column | Data Type | Key |
|---|---|---|
| order_id | INT | Primary Key |
| customer_id | INT | Foreign Key → customers |
| order_date | DATE | |
| status | VARCHAR(20) | |

### ORDER_ITEMS

| Column | Data Type | Key |
|---|---|---|
| item_id | INT | Primary Key |
| order_id | INT | Foreign Key → orders |
| product_id | INT | Foreign Key → products |
| quantity | INT | |
| unit_price | DECIMAL(10,2) | |

# Sample Data

### CUSTOMERS

| customer_id | customer_name | city |
|---|---|---|
| 1 | Rahul | Hyderabad |
| 2 | Priya | Vijayawada |
| 3 | Asha | Guntur |
| 4 | Kiran | Hyderabad |
| 5 | Ravi | Guntur |

### PRODUCTS

| product_id | product_name | category | price |
|---|---|---|---|
| 101 | Laptop | Electronics | 50000 |
| 102 | Mobile | Electronics | 20000 |
| 103 | Headphones | Accessories | 2000 |
| 104 | Keyboard | Accessories | 1500 |
| 105 | Mouse | Accessories | 800 |

### ORDERS

| order_id | customer_id | order_date | status |
|---|---|---|---|
| 1001 | 1 | 2026-01-10 | Delivered |
| 1002 | 2 | 2026-01-12 | Delivered |
| 1003 | 3 | 2026-02-05 | Pending |
| 1004 | 1 | 2026-02-15 | Delivered |
| 1005 | 4 | 2026-03-01 | Cancelled |

### ORDER ITEMS

| item_id | order_id | product_id | quantity | unit_price |
|---|---|---|---|---|
| 1 | 1001 | 101 | 1 | 50000 |
| 2 | 1001 | 103 | 2 | 2000 |
| 3 | 1002 | 102 | 1 | 20000 |
| 4 | 1003 | 104 | 2 | 1500 |
| 5 | 1004 | 103 | 1 | 2000 |
| 6 | 1004 | 105 | 2 | 800 |
| 7 | 1005 | 102 | 1 | 20000 |

# Queries

### 1. Display All Customers

```
SELECT * FROM customers;
```

**Output**

| customer_id | customer_name | city |
|---|---|---|
| 1 | Rahul | Hyderabad |
| 2 | Priya | Vijayawada |
| 3 | Asha | Guntur |
| 4 | Kiran | Hyderabad |
| 5 | Ravi | Guntur |

### 2. Display All Products

```
SELECT * FROM products;
```

**Output**

| product_id | product_name | category | price |
|---|---|---|---|
| 101 | Laptop | Electronics | 50000 |
| 102 | Mobile | Electronics | 20000 |
| 103 | Headphones | Accessories | 2000 |
| 104 | Keyboard | Accessories | 1500 |
| 105 | Mouse | Accessories | 800 |

### 3. Find Total Number of Orders

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

**Output**

| total_orders |
|---|
| 5 |

### 4. Calculate Total Sales

```
SELECT SUM(quantity * unit_price) AS total_sales
FROM order_items;
```

**Output**

| total_sales |
|---|
| 100600 |

### 5. Find Total Sales from Delivered Orders

```
SELECT SUM(oi.quantity * oi.unit_price) AS delivered_sales
FROM order_items oi
JOIN orders o ON oi.order_id = o.order_id
WHERE o.status = 'Delivered';
```

**Output**

| delivered_sales |
|---|
| 77600 |

### 6. Find the Most Expensive Product

```
SELECT *
FROM products
ORDER BY price DESC
LIMIT 1;
```

**Output**

| product_id | product_name | category | price |
|---|---|---|---|
| 101 | Laptop | Electronics | 50000 |

### 7. Find Customer Order Details

```
SELECT c.customer_name,
       o.order_id,
       o.order_date,
       o.status
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id;
```

**Output**

| customer_name | order_id | order_date | status |
|---|---|---|---|
| Rahul | 1001 | 2026-01-10 | Delivered |
| Priya | 1002 | 2026-01-12 | Delivered |
| Asha | 1003 | 2026-02-05 | Pending |
| Rahul | 1004 | 2026-02-15 | Delivered |
| Kiran | 1005 | 2026-03-01 | Cancelled |

### 8. Find Number of Orders by Status

```
SELECT status,
       COUNT(*) AS total_orders
FROM orders
GROUP BY status;
```

**Output**

| status | total_orders |
|---|---|
| Cancelled | 1 |
| Delivered | 3 |
| Pending | 1 |

### 9. Find Customers from Hyderabad

```
SELECT *
FROM customers
WHERE city = 'Hyderabad';
```

**Output**

| customer_id | customer_name | city |
|---|---|---|
| 1 | Rahul | Hyderabad |
| 4 | Kiran | Hyderabad |

### 10. Find Products Costing More Than 5000

```
SELECT *
FROM products
WHERE price > 5000;
```

**Output**

| product_id | product_name | category | price |
|---|---|---|---|
| 101 | Laptop | Electronics | 50000 |
| 102 | Mobile | Electronics | 20000 |

### 11. Find the Cheapest Product

```
SELECT *
FROM products
ORDER BY price ASC
LIMIT 1;
```

**Output**

| product_id | product_name | category | price |
|---|---|---|---|
| 105 | Mouse | Accessories | 800 |

### 12. Find Delivered Orders

```
SELECT *
FROM orders
WHERE status = 'Delivered';
```

**Output**

| order_id | customer_id | order_date | status |
|---|---|---|---|
| 1001 | 1 | 2026-01-10 | Delivered |
| 1002 | 2 | 2026-01-12 | Delivered |
| 1004 | 1 | 2026-02-15 | Delivered |

### 13. Find Products Priced Between 1000 and 25000

```
SELECT *
FROM products
WHERE price BETWEEN 1000 AND 25000;
```

**Output**

| product_id | product_name | category | price |
|---|---|---|---|
| 102 | Mobile | Electronics | 20000 |

| product_id | product_name | category | price |
|---|---|---|---|
| 103 | Headphones | Accessories | 2000 |
| 104 | Keyboard | Accessories | 1500 |

### 14. Find Customers Who Have Not Placed Orders

```
SELECT c.customer_id,
       c.customer_name
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

**Output**

| customer_id | customer_name |
|---|---|
| 5 | Ravi |

### 15. Calculate Total Quantity Sold

```
SELECT SUM(quantity) AS total_quantity_sold
FROM order_items;
```

**Output**

| total_quantity_sold |
|---|
| 10 |

### 16. Display All Customers and Their Orders (Including Customers Without Orders)

```
SELECT c.customer_name,
       o.order_id,
       o.status
FROM customers c
LEFT JOIN orders o ON c.customer_id = o.customer_id;
```

**Output**

| customer_name | order_id | status |
|---|---|---|
| Rahul | 1001 | Delivered |
| Rahul | 1004 | Delivered |
| Priya | 1002 | Delivered |
| Asha | 1003 | Pending |
| Kiran | 1005 | Cancelled |
| Ravi | NULL | NULL |

### 17. Find the Number of Products in Each Order

```
SELECT order_id,
       COUNT(DISTINCT product_id) AS total_products
FROM order_items
GROUP BY order_id;
```

**Output**

| order_id | total_products |
|---|---|
| 1001 | 2 |

| order_id | total_products |
|---|---|
| 1002 | 1 |
| 1003 | 1 |
| 1004 | 2 |
| 1005 | 1 |

### 18. Count Customers in Each City

```
SELECT city,
       COUNT(*) AS total_customers
FROM customers
GROUP BY city;
```

**Output**

| city | total_customers |
|---|---|
| Guntur | 2 |
| Hyderabad | 2 |
| Vijayawada | 1 |

### 19. Find the Average Price of Products

```
SELECT AVG(price) AS average_price
FROM products;
```

**Output**

| average_price |
|---|
| 14860 |

### 20. Display Complete Order Details

```
SELECT o.order_id,
       c.customer_name,
       p.product_name,
       oi.quantity,
       oi.unit_price,
       oi.quantity * oi.unit_price AS total_amount,
       o.status
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id;
```

**Output**

| order_id | customer_name | product_name | quantity | unit_price | total_amount | status |
|---|---|---|---|---|---|---|
| 1001 | Rahul | Laptop | 1 | 50000 | 50000 | Delivered |
| 1001 | Rahul | Headphones | 2 | 2000 | 4000 | Delivered |
| 1002 | Priya | Mobile | 1 | 20000 | 20000 | Delivered |
| 1003 | Asha | Keyboard | 2 | 1500 | 3000 | Pending |
| 1004 | Rahul | Headphones | 1 | 2000 | 2000 | Delivered |
| 1004 | Rahul | Mouse | 2 | 800 | 1600 | Delivered |
| 1005 | Kiran | Mobile | 1 | 20000 | 20000 | Cancelled |